The implicit BTCS scheme and the Thomas algorithm

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May 2026; revised October 2026

1 Setting and motivation

The explicit FTCS scheme of the previous note is conditionally stable: $\alpha = (\sigma^2/2)\Delta\tau/\Delta x^2 \leq 1/2$ forces $M \gtrsim N^2$ time steps and $O(N^3)$ total work. This note implements BTCS (Backward-Time, Centred-Space), an *unconditionally stable* scheme: any $\Delta\tau > 0$ produces a convergent solution, decoupling M from N.

The cost of unconditional stability is that each time step requires solving a linear system. This note has therefore two deliverables:

1. the BTCS pricer for European options;
2. the *Thomas algorithm*, a specialised Gaussian elimination that solves tridiagonal systems in $O(N)$ work.

Thomas is implemented from scratch as a standalone module (`quantlib.thomas` in Python, `quant::pde::thomas` in C++). It is reused by the Crank–Nicolson pricer through the generic θ-scheme stepper (`quantlib.theta_scheme`); the American pricer replaces it with PSOR (`quantlib.psor`).

2 The BTCS discretisation

We discretise the transformed Black–Scholes PDE

$$\frac{\partial V}{\partial \tau} = \frac{\sigma^2}{2}\frac{\partial^2 V}{\partial x^2} + \mu\,\frac{\partial V}{\partial x} - r\,V \tag{1}$$

on the grid of the finite-difference foundations note. BTCS evaluates every spatial stencil at the *new* time level $n+1$:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} = \frac{\sigma^2}{2}\,\frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta x^2} + \mu\,\frac{V_{j+1}^{n+1} - V_{j-1}^{n+1}}{2\Delta x} - r\,V_j^{n+1}.$$

Multiplying by $\Delta\tau$ and grouping all V^{n+1} on the left, V^n on the right:

$$\boxed{b_-\,V_{j-1}^{n+1} + b_0\,V_j^{n+1} + b_+\,V_{j+1}^{n+1} \;=\; V_j^n} \tag{2}$$

with

$$b_- = -\alpha + \frac{\mu\,\Delta\tau}{2\,\Delta x}, \qquad b_0 = 1 + 2\alpha + r\,\Delta\tau, \qquad b_+ = -\alpha - \frac{\mu\,\Delta\tau}{2\,\Delta x},$$

where $\alpha = (\sigma^2/2)\Delta\tau/\Delta x^2$ as before.

Equation (2) is the FTCS scheme *written from the other side*: where FTCS reads "V_j^{n+1} equals an explicit combination of V^n", BTCS reads "V_j^n equals an explicit combination of V^{n+1}". The latter is a system to be solved for the unknowns V^{n+1}. The two schemes are the $\theta = 0$ and $\theta = 1$ extremes of the family

$$b_-\,V_{j-1}^{n+1} + b_0\,V_j^{n+1} + b_+\,V_{j+1}^{n+1} = a_-\,V_{j-1}^n + a_0\,V_j^n + a_+\,V_{j+1}^n,$$

the θ-scheme, with Crank–Nicolson at $\theta = 1/2$ (see the Crank–Nicolson note).

3 The tridiagonal system

For each time step, equation (2) written for the interior nodes $j = 1, \dots, N-1$ is a linear system $A\,\mathbf{V}_{\text{int}}^{n+1} = \mathbf{d}$ where $A \in \mathbb{R}^{(N-1)\times(N-1)}$ is tridiagonal:

$$A = \begin{pmatrix} b_0 & b_+ & & & \\ b_- & b_0 & b_+ & & \\ & b_- & b_0 & b_+ & \\ & & \ddots & \ddots & \ddots \\ & & & b_- & b_0 \end{pmatrix}.$$

The boundary nodes V_0^{n+1} and V_N^{n+1} are known from the Dirichlet conditions of the foundations note; they appear in the right-hand side:

$$\mathbf{d} = \begin{pmatrix} V_1^n - b_-\,V_0^{n+1} \\ V_2^n \\ \vdots \\ V_{N-2}^n \\ V_{N-1}^n - b_+\,V_N^{n+1} \end{pmatrix}. \tag{3}$$

The first and last rows pick up the contributions from the boundary; the rest of $\mathbf{d}$ is just V^n at the interior.

Constancy of A in time. The coefficients b_-, b_0, b_+ are independent of n and j thanks to the constant-coefficient form (1). The matrix A is therefore the same at every time step; only the right-hand side $\mathbf{d}$ changes. This is the structural pay-off of the logarithmic transformation that we already exploited in FTCS, now in a stronger form: at each step we solve a system with the *same* matrix and a different right-hand side. The forward sweep of the Thomas algorithm can be done *once* at setup and reused, halving the per-step cost.

4 Stability: unconditional

The von Neumann analysis of (2) mirrors the foundations note, §9.3, extended with the convective and reactive terms. Substituting $V_j^n = g^n e^{ij\theta}$ and solving for g:

$$g(\theta) = \frac{1}{1 + 4\alpha \sin^2(\theta/2) - i\,\beta \sin\theta + r\,\Delta\tau}, \qquad \beta = \frac{\mu\,\Delta\tau}{\Delta x}. \tag{4}$$

The squared modulus,

$$|g|^2 = \frac{1}{\big(1 + 4\alpha \sin^2(\theta/2) + r\Delta\tau\big)^2 + \beta^2 \sin^2\theta},$$

has denominator ≥ 1 for every θ (every term is non-negative; one of them, 1, is strictly so). Hence $|g(\theta)| \leq 1$ unconditionally: BTCS is stable for any $\Delta\tau > 0$ and $\Delta x > 0$.

Combined with consistency ($O(\Delta\tau)+O(\Delta x^2)$ local truncation), the Lax equivalence theorem gives convergence:

$$\|V_h - V\|_\infty = O(\Delta\tau) + O(\Delta x^2).$$

Remark 4.1 (Stability is not monotonicity)**.** *L^2 stability does not imply L^∞ monotonicity. For very large $\Delta\tau$ at moderate Δx, the matrix A ceases to be an M-matrix: A^{-1} has negative entries, and the discrete operator can produce localised oscillations near the strike kink. These oscillations are bounded (stability holds), but they degrade local accuracy. For our parameter range this regime is far away; when we reach it (large-$\Delta\tau$ tests in the validation), we will see prices that are merely imprecise, never explosive.*

5 The Thomas algorithm

Solving $A\,\mathbf{x} = \mathbf{d}$ with general Gaussian elimination costs $O(N^3)$ — prohibitive at $N = 200$, hopeless at $N = 1000$. The Thomas algorithm exploits the tridiagonal structure to compute the same solution in $O(N)$ work, which is asymptotically optimal: just reading the input takes $O(N)$ time.

Setup. Consider the general tridiagonal system

$$\begin{pmatrix} \tilde{b}_1 & c_1 & & & \\ a_2 & \tilde{b}_2 & c_2 & & \\ & a_3 & \tilde{b}_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & \tilde{b}_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix}.$$

We write $\tilde{b}$ for the diagonal of the general system to distinguish from the BTCS-specific b_0, b_-, b_+.

Forward sweep. Standard Gaussian elimination, applied naively to a tridiagonal system, produces *no fill*: at each step the only non-zero entries below the pivot are a_i, and the elimination touches $\tilde{b}_i$ and d_i but not the rows below. The recursion is

$$\begin{aligned} c'_1 &= \frac{c_1}{\tilde{b}_1}, & d'_1 &= \frac{d_1}{\tilde{b}_1}, \\ m_i &= \tilde{b}_i - a_i\, c'_{i-1}, & c'_i &= \frac{c_i}{m_i}, & d'_i &= \frac{d_i - a_i\, d'_{i-1}}{m_i}, \quad i = 2, \ldots, n. \end{aligned} \tag{5}$$

The m_i are the *effective pivots*: what remains on the diagonal after eliminating a_i. After the sweep the system is upper bidiagonal:

$$\begin{pmatrix} 1 & c'_1 & & & \\ & 1 & c'_2 & & \\ & & 1 & c'_3 & \\ & & & \ddots & \end{pmatrix} \mathbf{x} = \mathbf{d}'.$$

Backward substitution. Trivial:

$$x_n = d'_n, \qquad x_i = d'_i - c'_i\, x_{i+1}, \quad i = n-1, n-2, \ldots, 1. \tag{6}$$

Operation count. The forward sweep does roughly $5n$ floating-point operations (one division for c'_i, three operations for d'_i, one for m_i). The backward substitution does $\sim 2n$. Total: $O(n)$ with constant ≈ 7.

Memory. Only four vectors of length $\leq n$: the three diagonals (modified in place if desired) and the right-hand side. No fill-in, no $O(n^2)$ storage.

Pre-factoring for repeated solves. The forward sweep (5) computes two distinct families of quantities: those depending only on $(a, \tilde{b}, c)$, namely $\{m_i, c'_i\}$, and those depending also on d, namely $\{d'_i\}$. When the matrix is constant across many right-hand sides — exactly our case, with one fixed A and M different right-hand sides — we compute the (m_i, c'_i) once at setup, and each subsequent solve costs only the d' recursion plus the backward substitution. The savings are $O(n)$ divisions per solve, i.e., a factor ~ 1.5–2 in practice.

6 Stability of Thomas without pivoting

The Thomas algorithm as written does *not* pivot: there is no provision for $m_i = 0$ or $|m_i|$ small. In general, Gaussian elimination without pivoting is unsafe — a small pivot amplifies round-off catastrophically. For Thomas there is a clean theorem that rules out the failure mode for our matrices.

Theorem 6.1 (Thomas without pivoting under diagonal dominance)**.** *If the tridiagonal matrix is diagonally dominant in the sense*

$$|\tilde{b}_i| \geq |a_i| + |c_i|, \quad i = 1, \ldots, n, \tag{7}$$

with strict inequality in at least one row, then all effective pivots m_i in (5) satisfy $|m_i| \geq |\tilde{b}_i| - |a_i| > 0$, and the algorithm is backward stable: the computed solution $\hat{\mathbf{x}}$ satisfies $(A + \delta A)\hat{\mathbf{x}} = \mathbf{d}$ with $\|\delta A\|_\infty \leq c_n\, \varepsilon_{\text{mach}} \|A\|_\infty$ for a small constant c_n proportional to n.

(Proof in Higham, *Accuracy and Stability of Numerical Algorithms*, §9.5.)

Application to BTCS. The BTCS matrix has $\tilde{b}_i = b_0$, $a_i = b_-$, $c_i = b_+$, all constant in i. Diagonal dominance reduces to

$$|b_0| \;\geq\; |b_-| + |b_+|.$$

We have $|b_0| = 1 + 2\alpha + r\Delta\tau$ (positive). Computing the right-hand side:

$$|b_-| + |b_+| = \left|\alpha - \frac{\mu\Delta\tau}{2\Delta x}\right| + \left|\alpha + \frac{\mu\Delta\tau}{2\Delta x}\right|.$$

When $\alpha \geq \left|\mu\Delta\tau/(2\Delta x)\right|$ — i.e., $\left|\mu\right|\Delta x/\sigma^2 \leq 1$, the grid Péclet condition of the FTCS note — both terms inside the absolute values have α dominating and the same sign, and we get $|b_-| + |b_+| = 2\alpha$. Then

$$|b_0| - (|b_-| + |b_+|) = 1 + r\Delta\tau > 0,$$

strict diagonal dominance. So Thomas without pivoting is provably stable for BTCS whenever $\mathrm{Pe}_h \leq 1$, which the project's parameter range satisfies with comfortable margin ($\mathrm{Pe}_h \leq 0.25$).

7 Convergence rates and computational cost

The BTCS error is $O(\Delta\tau) + O(\Delta x^2)$. With M now independent of N, the user has a choice:

Quadratic-precision regime ($M \sim N^2$). Choose $\Delta\tau \sim \Delta x^2$. Both error contributions are $O(\Delta x^2)$; the total error is $O(\Delta x^2)$. The computational cost is $O(N)\cdot M \cdot 2 = O(N \cdot N^2) = O(N^3)$, with the constant a factor ~ 7 from the Thomas count. *Same asymptotic cost as FTCS for the same precision.*

Linear-precision regime ($M \sim N$). Choose $\Delta\tau \sim \Delta x$. The error is dominated by $O(\Delta\tau) = O(\Delta x)$; the spatial $O(\Delta x^2)$ contribution is sub-dominant. The total error is $O(\Delta x)$. Cost: $O(N) \cdot M = O(N^2)$. *Acceptable when first-order precision is enough* (prototyping, calibration of insensitive parameters, sanity checks).

Crank–Nicolson preview. Crank–Nicolson achieves $O(\Delta\tau^2) + O(\Delta x^2)$ local truncation, so $M \sim N$ gives full $O(\Delta x^2)$ precision at $O(N^2)$ cost. That is the point where the explicit-vs-implicit trade decisively tips toward implicit.

Summary table.

Scheme	Local error	Stability	For $O(\Delta x^2)$	For $O(\Delta x)$
FTCS	$\Delta\tau + \Delta x^2$	$\alpha \leq 1/2$	$O(N^3)$ (forced)	not relevant
BTCS	$\Delta\tau + \Delta x^2$	unconditional	$O(N^3)$ ($\mathrm{M} \sim N^2$)	$O(N^2)$ ($\mathrm{M} \sim N$)
CN	$\Delta\tau^2 + \Delta x^2$	unconditional	$O(N^2)$ ($\mathrm{M} \sim N$)	overkill

8 Implementation outline

The deliverables are two modules each in Python and C++:

- `quantlib.thomas` (Python) / `quant::pde::thomas_*` (C++) provides `thomas_solve(sub, diag, sup, rhs)` for one-shot solves and the pair `thomas_factor(sub, diag, sup)` + `thomas_solve_factored(factor, rhs)` for repeated solves with a fixed matrix.
- `quantlib.btcs` (Python) / `quant::pde::btcs_*` (C++) provides `btcs_european_call/put` with the same signature as the FTCS pricers, except that there is no CFL check: any $M \geq 1$ is valid input.

The validation suite checks five properties:

1. Cross-validation of `thomas_solve` against `numpy.linalg.solve` (Python) and against an independent dense LU implementation in the C++ tests, for randomly-generated diagonally dominant matrices.
2. Cross-validation of BTCS against Black–Scholes closed-form, with tolerance scaled to the discretisation error.
3. Convergence rate at $\Delta\tau \sim \Delta x^2$: error reduction by a factor of ~ 4 when $N \to 2N$, $M \to 4M$.
4. First-order convergence in time at fixed (large) N, $M \to 2M$: error reduction by ~ 2.
5. *Large-$\Delta\tau$ stability test*: BTCS at $M = 10$ for $T = 1$ produces a finite, sensible price (although imprecise), with no explosion. This is the operational signature of unconditional stability.

References

References

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